Project:- House price prediction

Dataset Selection

Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Upload Dataset

```
from google.colab import files
uploaded = files.upload()
```

Choose Files house_price_50k.csv

house_price_50k.csv(text/csv) - 4229798 bytes, last modified: 4/20/2026 - 100% done

```
Saving house_price_50k.csv to house_price_50k.csv
```

Then Load Dataset

```
import pandas as pd

df = pd.read_csv("house_price_50k.csv")
df.head()
```

	area	bedrooms	bathrooms	floors	age	distance	garage	parking	garden	sec
0	1360	6	2	2	9	10	0	0	1	
1	4272	5	2	3	24	8	1	0	0	
2	3592	1	4	3	4	20	0	1	1	
3	966	5	2	2	6	14	1	0	0	
4	4926	4	3	1	18	9	0	0	0	

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Understanding the Data

```
df.shape
```

```
(50000, 19)
```

```
df.columns
```

```
Index(['area', 'bedrooms', 'bathrooms', 'floors', 'age', 'distance', 'garage',
       'parking', 'garden', 'security', 'school_nearby', 'hospital_nearby',
       'shopping_mall_nearby', 'public_transport', 'crime_rate',
       'population_density', 'location', 'income_level', 'price'],
      dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 19 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   area                  50000 non-null  int64
 1   bedrooms              50000 non-null  int64
 2   bathrooms             50000 non-null  int64
 3   floors                50000 non-null  int64
 4   age                   50000 non-null  int64
 5   distance              50000 non-null  int64
 6   garage                50000 non-null  int64
 7   parking               50000 non-null  int64
 8   garden                50000 non-null  int64
 9   security              50000 non-null  int64
 10  school_nearby         50000 non-null  int64
 11  hospital_nearby       50000 non-null  int64
 12  shopping_mall_nearby  50000 non-null  int64
 13  public_transport      50000 non-null  int64
 14  crime_rate            50000 non-null  float64
 15  population_density    50000 non-null  int64
 16  location              50000 non-null  object
 17  income_level          50000 non-null  object
 18  price                 50000 non-null  float64
dtypes: float64(2), int64(15), object(2)
memory usage: 7.2+ MB
```

```
df.describe
```

```
pandas.core.generic.NDFrame.describe
def describe(percentiles=None, include=None, exclude=None) -> Self
```

```
/usr/local/lib/python3.12/dist-packages/pandas/core/generic.py
Generate descriptive statistics.

Descriptive statistics include those that summarize the central
tendency, dispersion and shape of a
dataset's distribution, excluding ``NaN`` values.
```

Check Missing Values

```
df.isnull().sum()
```

	0
area	0
bedrooms	0
bathrooms	0
floors	0
age	0
distance	0
garage	0
parking	0
garden	0
security	0
school_nearby	0
hospital_nearby	0
shopping_mall_nearby	0
public_transport	0
crime_rate	0
population_density	0
location	0
income_level	0
price	0

dtype: int64

After checking missing values using df.isnull().sum(), it was observed that all columns contain 0 missing values. Hence, no data cleaning was required for missing values.

Check Duplicate Data

```
df.duplicated().sum()
```

```
np.int64(0)
```

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = le.fit_transform(df[col])

df.head()
```

	area	bedrooms	bathrooms	floors	age	distance	garage	parking	garden	sec
0	1360	6	2	2	9	10	0	0	1	
1	4272	5	2	3	24	8	1	0	0	
2	3592	1	4	3	4	20	0	1	1	
3	966	5	2	2	6	14	1	0	0	
4	4926	4	3	1	18	9	0	0	0	

Next steps: Generate code with df | New interactive sheet

Final Conclusion

The dataset was successfully cleaned and preprocessed. It contains no missing or duplicate values, and all necessary transformations were applied. The dataset is now ready for model building and prediction.